TMJC 2026 H2 MATH (9758/01) JC 2 PRELIMINARY EXAMINATION

Qn	Solution
1	**Implicit Differentiation, Normal**
(a)	$2x^2 + xy + y^2 = 14$ Differentiating w.r.t. x: $4x + x\frac{\mathrm{d}y}{\mathrm{d}x} + y + 2y\frac{\mathrm{d}y}{\mathrm{d}x} = 0$ $\frac{\mathrm{d}y}{\mathrm{d}x}(x + 2y) = -4x - y$ $\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{4x + y}{x + 2y}$ (shown)
(b)	Since gradient of L is undefined at $P \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = 0$ at P $\therefore y = -4x$ Substituting $y = -4x$ into $2x^2 + xy + y^2 = 14$ $2x^2 + x(-4x) + (-4x)^2 = 14$ $x^2 = 1 \Rightarrow x = 1$ or $x = -1$ Since P is in the second quadrant, $x = -1$ and $y = 4$ Coordinates of P $(-1, 4)$.
(c)	$\therefore$ Equation of L at P is $x = -1$ Substituting $x = -1$ into $2x^2 + xy + y^2 = 14$, We get $y^2 - y - 12 = 0$ Using GC, $y = 4$ or $y = -3$ (reject $y = 4$ which represents pt P) Coordinates of Q $(-1, -3)$.

Qn	Solution
2	**Integration Techniques**
(a)	$u = x^3 + 1 \quad \Rightarrow \dfrac{\mathrm{d}u}{\mathrm{d}x} = 3x^2$ $\Rightarrow x^2 \mathrm{d}x = \dfrac{1}{3}\mathrm{d}u$ When $x = 2$, $u = 9$ When $x = 0$, $u = 1$ $\displaystyle\int_0^2 \frac{x^5}{\sqrt{x^3+1}}\mathrm{d}x$ $\displaystyle= \int_0^2 \frac{x^3}{\sqrt{x^3+1}}\left(x^2\right)\mathrm{d}x$ $\displaystyle= \int_1^9 \frac{u-1}{\sqrt{u}}\left(\frac{1}{3}\right)\mathrm{d}u$ $\displaystyle= \frac{1}{3}\int_1^9 \sqrt{u} - \frac{1}{\sqrt{u}}\mathrm{d}u$ $\displaystyle= \frac{1}{3}\left[\frac{2}{3}u^{\frac{3}{2}} - 2\sqrt{u}\right]_1^9$ $\displaystyle= \frac{1}{3}\left[12 + \frac{4}{3}\right] = \frac{40}{9}$
(b)	$\displaystyle\int \sin^{-1} x \,\mathrm{d}x$ $\displaystyle= x\sin^{-1}x - \int x\left[\frac{1}{\sqrt{1-x^2}}\right]\mathrm{d}x$ $u = \sin^{-1}x \qquad v = 1$ $\dfrac{\mathrm{d}u}{\mathrm{d}x} = \dfrac{1}{\sqrt{1-x^2}} \qquad \int v\,\mathrm{d}x = x$ $\displaystyle= x\sin^{-1}x + \frac{1}{2}\int \frac{-2x}{\sqrt{1-x^2}}\mathrm{d}x$ $\mathrm{f}(x) = 1-x^2$ $\mathrm{f}'(x) = -2x$ $\displaystyle= x\sin^{-1}x + \left(\frac{1}{2}\right)\frac{\sqrt{1-x^2}}{\left(\frac{1}{2}\right)} + c$ $= x\sin^{-1}x + \sqrt{1-x^2} + c$

Qn	Solution
3	**Differential Equations**

$\left(1+x^2\right)\dfrac{\mathrm{d}y}{\mathrm{d}x} = y\left(1+y^2\right) - - - (1)$

$y = \tan\theta \Rightarrow 1+y^2 = \sec^2\theta - - - (2)$

$\dfrac{\mathrm{d}y}{\mathrm{d}x} = \sec^2\theta \dfrac{\mathrm{d}\theta}{\mathrm{d}x} - - - (3)$

Sub (2) and (3) into (1):

$\left(1+x^2\right)\sec^2\theta \dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \tan\theta\sec^2\theta$

$\dfrac{\mathrm{d}\theta}{\mathrm{d}x} = \dfrac{\tan\theta}{1+x^2}$ (shown)

$\int \cot\theta \, \mathrm{d}\theta = \int \dfrac{1}{1+x^2} \, \mathrm{d}x$

$\ln\left|\sin\theta\right| = \tan^{-1}x + c$

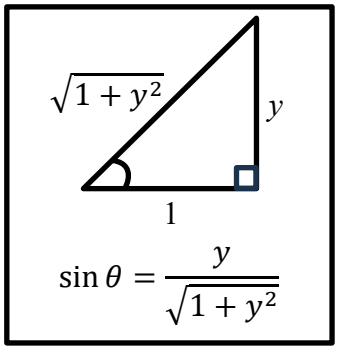
Since $0 < \theta < \dfrac{\pi}{2}$, $\ln\left(\sin\theta\right) = \tan^{-1}x + c$

$y = \tan\theta \Rightarrow \sin\theta = \dfrac{y}{\sqrt{1+y^2}}$

$\ln\left(\dfrac{y}{\sqrt{1+y^2}}\right) = \tan^{-1}x + c$

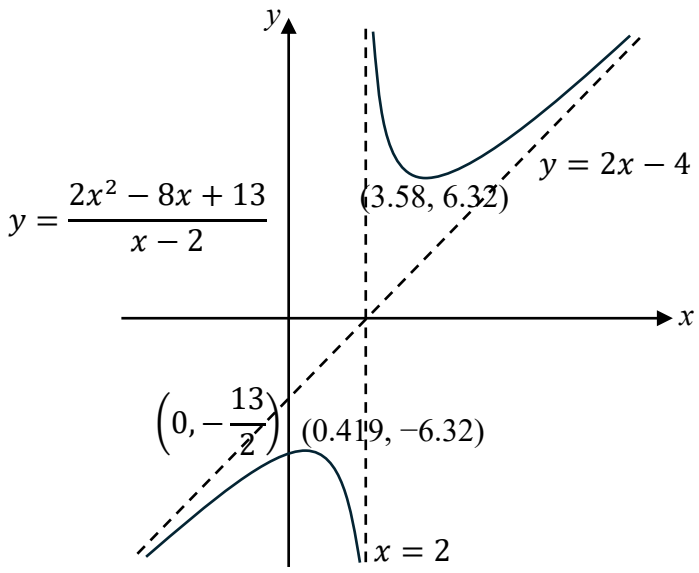
When $x = 0, y = 1, \therefore c = -\dfrac{1}{2}\ln 2$

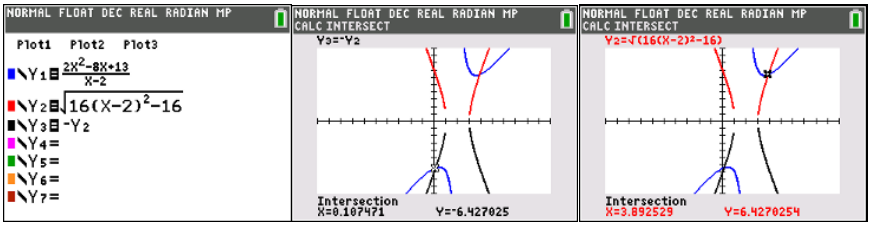
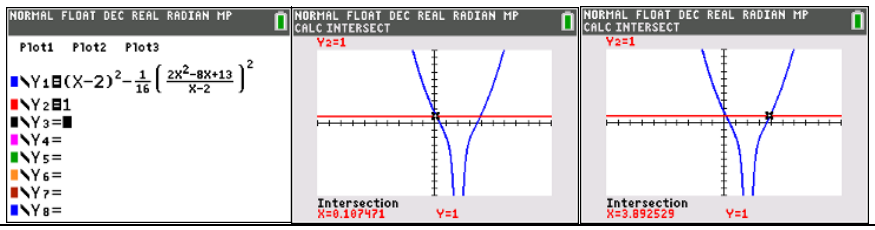
$\ln\left(\dfrac{y}{\sqrt{1+y^2}}\right) = \tan^{-1}x - \dfrac{1}{2}\ln 2.$

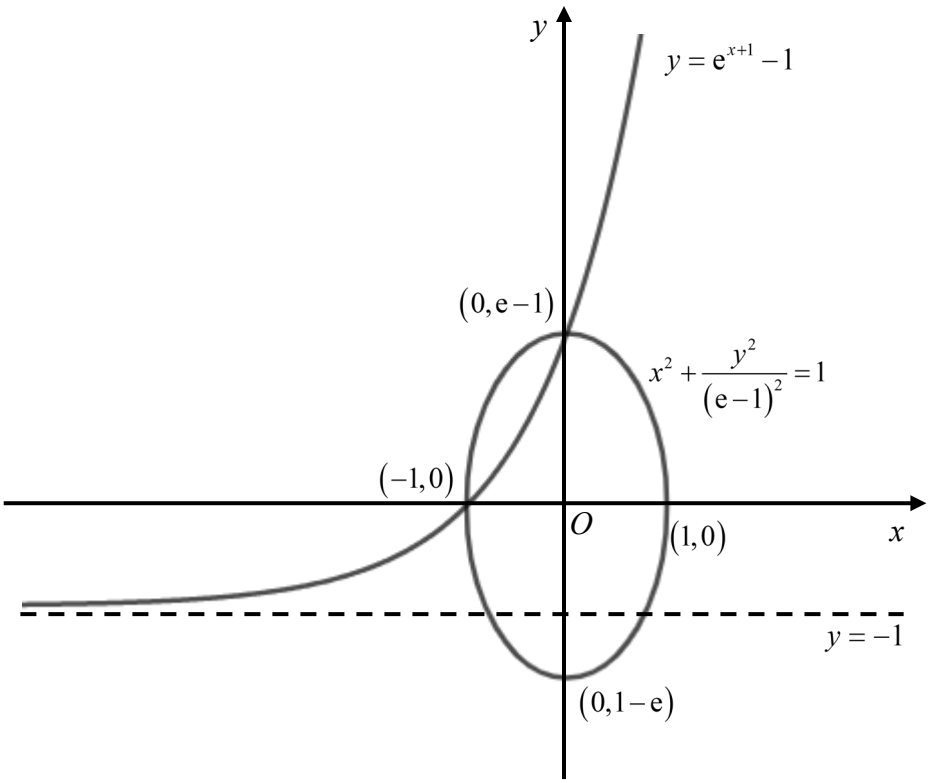


$\sin\theta = \dfrac{y}{\sqrt{1+y^2}}$

Qn	Solution
4	**Summation and Definite Integral**
(a)	$\sum_{r=4}^{n}\left[(r-3)(2r-1)\right]$ $=\sum_{r=4}^{n}\left(2r^2-7r+3\right)$ $=2\sum_{r=4}^{n}r^2+\sum_{r=4}^{n}(3-7r)$ $=2\left(\sum_{r=1}^{n}r^2-\sum_{r=1}^{3}r^2\right)+\frac{n-3}{2}(-25+3-7n)$ $=2\left(\frac{n}{6}(n+1)(2n+1)-14\right)-\frac{n-3}{2}(22+7n)$ $=\frac{n}{3}(n+1)(2n+1)-\frac{1}{2}(n-3)(22+7n)-28$
(b)(i)	$S_n=1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{n}$
(b)(ii)	$S=1+\frac{1}{2}+\frac{1}{3}+...$ $>\lim_{n\to\infty}\int_1^{n+1}\frac{1}{x}\,\mathrm{d}x$ $=\lim_{n\to\infty}\left[\ln\lvert x\rvert\right]_1^{n+1}$ $=\lim_{n\to\infty}\ln(n+1)$ As $n\to\infty, \ln(n+1)\to\infty$ $\therefore S=1+\frac{1}{2}+\frac{1}{3}+...$ is divergent.

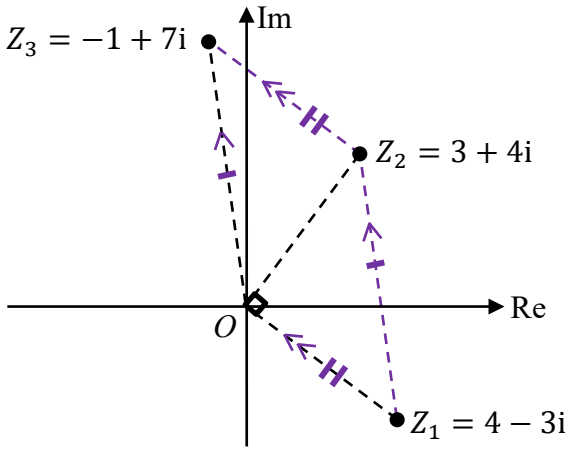
Qn	Solution
5	**Graphing Techniques**
(a)	$y=\dfrac{2x^2+kx+13}{x-2}=2x-4+\dfrac{A}{x-2}$ $=\dfrac{(2x-4)(x-2)+A}{x-2}$ $=\dfrac{2x^2-8x+8+A}{x-2}$ Comparing expressions, $k=-8$ (shown) **Alternative** $\begin{array}{r} 2x-4 \\ x-2\overline{)\ 2x^2+kx\ +13} \\ \underline{-(2x^2-4x)} \\ (k+4)x+13 \\ \underline{-(-4x+8)} \\ (k+8)x+5 \end{array}$ $\therefore k+8=0$ $k=-8$ (shown)
(b)	
(c)	$\dfrac{(x-2)^2}{a^2}-\dfrac{1}{b^2}\left(\dfrac{2x^2-8x+13}{x-2}\right)^2=1$ refers to the intersection between curve C and hyperbola with equation $\dfrac{(x-2)^2}{a^2}-\dfrac{y^2}{b^2}=1$. Finding oblique asymptotes of hyperbola,

Qn	Solution
	$\frac{(x-2)^2}{a^2}-\frac{y^2}{b^2}=0$ $\frac{y^2}{b^2}=\frac{(x-2)^2}{a^2}$ $\frac{y}{b}=\pm\frac{x-2}{a}$ $y=\pm\frac{b}{a}(x-2)$ Comparing asymptotes of hyperbola and oblique asymptote of curve C, to have 2 intersections (2 real roots) between the graphs, gradient of the asymptote with the positive gradient must be more than 2. $\therefore \frac{b}{a}>2$ $b>2a$
(d)	For $a=1$ and $b=4$, equation of hyperbola $(x-2)^2-\frac{y^2}{16}=1$. $(x-2)^2-\frac{y^2}{16}=1$ $y=\pm\sqrt{16(x-2)^2-16}$ Using GC, $x=0.107$ or $x=3.89$ (3 s.f.)  **Alternative** For $a=1$ and $b=4$, $(x-2)^2-\frac{1}{16}\left(\frac{2x^2-8x+13}{x-2}\right)^2=1$ Using GC, $x=0.107$ or $x=3.89$ (3 s.f.) 

Qn	Solution
6	**Graphing Techniques + Definite Integral**
(a)	
(b)	Volume $= (\mathrm{e}-1)^2 \pi \int_{-1}^{0} (1-x^2)\, \mathrm{d}x - \pi \int_{-1}^{0} (\mathrm{e}^{x+1} - 1)^2\, \mathrm{d}x$ $= (\mathrm{e}-1)^2 \pi \int_{-1}^{0} (1-x^2)\, \mathrm{d}x - \pi \int_{-1}^{0} (\mathrm{e}^{2x+2} - 2\mathrm{e}^{x+1} + 1)\, \mathrm{d}x$ $= (\mathrm{e}-1)^2 \pi \left[x - \frac{1}{3}x^3 \right]_{-1}^{0} - \pi \left[\frac{1}{2}\mathrm{e}^{2x+2} - 2\mathrm{e}^{x+1} + x \right]_{-1}^{0}$ $= (\mathrm{e}-1)^2 \pi \left(-(-1) + \frac{1}{3}(-1)^3 \right) - \pi \left(\frac{1}{2}\mathrm{e}^2 - 2\mathrm{e} - \frac{1}{2} + 2 + 1 \right)$ $= \frac{2}{3}(\mathrm{e}^2 - 2\mathrm{e} + 1)\pi - \left(\frac{1}{2}\mathrm{e}^2 - 2\mathrm{e} + \frac{5}{2} \right)\pi$ $= \left(\frac{1}{6}\mathrm{e}^2 + \frac{2}{3}\mathrm{e} - \frac{11}{6} \right)\pi$

Qn	Solution
7	**Maclaurin Series (Small Angle Approximation)**
(a)	By Sine Rule, $\frac{AC}{\sin\left(\frac{\pi}{3}\right)} = \frac{2}{\sin\left(\pi - \frac{\pi}{3} - \theta\right)}$ $\frac{AC}{\frac{\sqrt{3}}{2}} = \frac{2}{\sin\left(\frac{2\pi}{3} - \theta\right)}$ $AC = \frac{\sqrt{3}}{\sin\left(\frac{2\pi}{3} - \theta\right)}$ $AC = \frac{\sqrt{3}}{\sin\left(\frac{2\pi}{3}\right)\cos\theta - \cos\left(\frac{2\pi}{3}\right)\sin\theta}$ $AC = \frac{\sqrt{3}}{\frac{\sqrt{3}}{2}\cos\theta - \left(-\frac{1}{2}\right)\sin\theta}$ $AC = \frac{2\sqrt{3}}{\sqrt{3}\cos\theta + \sin\theta}$ (Shown)
(b)	For small θ, $AC \approx \frac{2\sqrt{3}}{\sqrt{3}\left(1 - \frac{\theta^2}{2} + ...\right) + (\theta + ...)}$ $AC = \frac{2\sqrt{3}}{\sqrt{3}\left[\left(1 - \frac{\theta^2}{2} + ...\right) + \frac{1}{\sqrt{3}}(\theta + ...)\right]}$ $\approx 2\left[1 + \left(\frac{\theta}{\sqrt{3}} - \frac{\theta^2}{2}\right)\right]^{-1}$ $= 2\left[1 + (-1)\left(\frac{\theta}{\sqrt{3}} - \frac{\theta^2}{2}\right) + \frac{(-1)(-2)}{2!}\left(\frac{\theta}{\sqrt{3}} - \frac{\theta^2}{2}\right)^2 + \cdots\right]$ $= 2\left(1 - \frac{\theta}{\sqrt{3}} + \frac{\theta^2}{2} + \frac{\theta^2}{3} + \cdots\right) \approx 2 - \frac{2}{\sqrt{3}}\theta + \frac{5}{3}\theta^2$
(c)	Expansion valid for $\left\lvert\frac{\theta}{\sqrt{3}} - \frac{\theta^2}{2}\right\rvert < 1$ $-0.95017 < \theta < 2.10487$ $-0.950 < \theta < 2.10$ (3 s.f) **Note:** If context is considered, $0 < \theta < \frac{2\pi}{3}$

8	**Complex Numbers**
(a)	Since $z_1 = 4 + k\text{i}$ is a root, $\Rightarrow (4+k\text{i})^2 - (7+\text{i})(4+k\text{i}) + (24+7\text{i}) = 0$ $\Rightarrow (16 + 8k\text{i} + k^2\text{i}^2) - (28 + 7k\text{i} + 4\text{i} + k\text{i}^2) + (24+7\text{i}) = 0$ $\Rightarrow (16 - k^2 - 28 + k + 24) + (8k - 7k - 4 + 7)\text{i} = 0$ Comparing imaginary part, $k + 3 = 0 \quad \Rightarrow k = -3$ (shown) $z^2 - (7+\text{i})z + (24+7\text{i}) = [z - (4-3\text{i})][z - z_2]$ $= z^2 - z_2 z - (4-3\text{i})z + (4-3\text{i})z_2$ $= z^2 - (z_2 + (4-3\text{i}))z + (4-3\text{i})z_2$ Comparing coefficient of z: $7 + \text{i} = z_2 + (4-3\text{i}) \quad \Rightarrow z_2 = 3 + 4\text{i}$ **Alternatively,** $24 + 7\text{i} = (4-3\text{i})z_2$ Comparing constant: $\Rightarrow z_2 = \dfrac{(24+7\text{i})(4+3\text{i})}{(4-3\text{i})(4+3\text{i})} = \dfrac{96 + 100\text{i} - 21}{25} = \dfrac{75+100\text{i}}{25} = 3 + 4\text{i}$ **<u>Method 2</u>** Let other root, $z_2 = x + y\text{i}$ $[z - (4+k\text{i})][z - (x+y\text{i})]$ $= z^2 - (x+y\text{i})z - (4+k\text{i})z + (4+k\text{i})(x+y\text{i})$ $= z^2 - [(x+4) + (y+k)\text{i}]z + [(4x - ky) + (4y + kx)\text{i}]$ $= z^2 - (7+\text{i})z + (24+7\text{i})$ Comparing real part of coefficient of z: $x + 4 = 7 \quad \Rightarrow x = 3$ Comparing imaginary part of coefficient of z: $y + k = 1 \quad ---(1)$ Comparing imaginary part of constant $4y + 3k = 7 \quad ---(2)$ $(2) - 3(1) : y = 4 \qquad \Rightarrow k = -3$ (shown) $\therefore z_2 = 3 + 4\text{i}$
(b)	$z_2 = 3 + 4\text{i} = \text{i}(4 - 3\text{i}) = \text{i}z_1$ The point Z_2 is a 90° **anti-clockwise** rotation **about the origin** of the point Z_1. **<u>Alternative:</u>** The point Z_1 is a 90° **clockwise** rotation about the origin of the point Z_2.

(c)	
(d)	$OZ_1Z_2Z_3$ is a parallelogram
(e)	Smallest possible n is 5 $(z_1)^2 = (4-3\text{i})^2 = 7-24\text{i}$ Since the polynomial has all real coefficients, non-real roots occur in conjugate pairs, so there are at least 5 roots (one real root, and 2 complex conjugate pairs)

Qn 9 — Arithmetic and Geometric Series

(a)(i)

$$A_n = \frac{n}{2}\left[2(1500)+(n-1)(20)\right]+bn$$
$$= 1500n + n(n-1)(10) + bn$$
$$= 1500n + 10n^2 - 10n + bn$$
$$= 10n^2 + (1490+b)n \text{ (shown)}$$

(a)(ii)

Year	Start of Year (Jan)	End of Year (Dec)
2026 $n=1$	1500	$1500(1.03)$
$n=2$	$1500(1.03)+1500$	$1500(1.03)^2+1500(1.03)$
$n=3$	$1500(1.03)^2+1500(1.03)+1500$	$1500(1.03)^3+1500(1.03)^2$ $+1500(1.03)$
		….

$$B_n = 1500(1.03)^n + 1500(1.03)^{n-1} + 1500(1.03)^{n-2} \ldots + 1500(1.03)$$
$$= 1500(1.03)\left(1+1.03+1.03^2+\ldots+(1.03)^{n-1}\right)$$
$$= 1500(1.03)\left(\frac{1-(1.03)^n}{1-1.03}\right)$$
$$= -51500\left(1-(1.03)^n\right)$$
$$= 51500\left(1.03^n-1\right) \text{ (shown)}$$

Or

$$= 1500(1.03)\left(\frac{(1.03)^n-1}{1.03-1}\right)$$
$$= 51500\left(1.03^n-1\right) \text{ (shown)}$$

(b)

$A_5 = 9200$

$10(5)^2 + (1490+b)5 = 9200$

$b = 300$

(c)

Using $b = 300$, $A_n = 10n^2 + 1790n$

$B_n = 51500\left(1.03^n - 1\right)$

Using GC,

When $n = 16$, $B_n = 31142.38 < A_n = 31200$

When $n = 17$, $B_n = 33621.65 > A_n = 33320$

Minimum n = 17

Hence Bob's saving will exceed Aaron's saving on 31 Dec 2042.

Difference in their total amount = 33621.65 – 33320 = \$301.65

(d)

Interest rate is now x %

$A_8 = 10(8)^2 + 1790(8) = 14960$

Want to solve $B_8 > 14960$

$$B_8 = 1500\left(1+\frac{x}{100}\right)\left(\frac{1-\left(1+\frac{x}{100}\right)^8}{1-\left(1+\frac{x}{100}\right)}\right) > 14960$$

Using GC, $x > 4.8824$

Minimum
$x = 4.89$ (2 d.p)

Qn	Solution
10	**3D Vectors**
(a)	$p_1 : \mathbf{r} \bullet \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 5$

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} \alpha \\ -2 \\ 1 \end{pmatrix}, \alpha, \lambda \in \mathbb{R}, \alpha \neq 0$$

$$\sin\left(\sin^{-1}\frac{4}{9}\right) = \frac{\left|\begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \bullet \begin{pmatrix} \alpha \\ -2 \\ 1 \end{pmatrix}\right|}{\sqrt{9}\sqrt{\alpha^2+5}}$$

$$\frac{4}{9} = \frac{|\alpha+6|}{3\sqrt{\alpha^2+5}}$$

$$12\sqrt{\alpha^2+5} = 9|\alpha+6|$$

$$\left(12\sqrt{\alpha^2+5}\right)^2 = \left(9|\alpha+6|\right)^2$$

$$144\left(\alpha^2+5\right) = 81(\alpha+6)^2$$

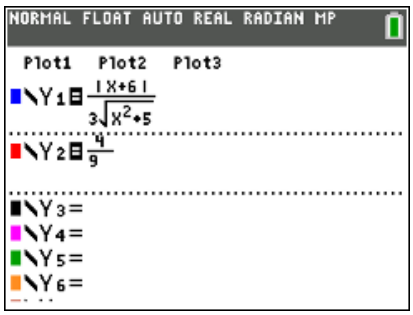
$$\frac{7}{9}\alpha^2 - 12\alpha - \frac{244}{9} = 0$$

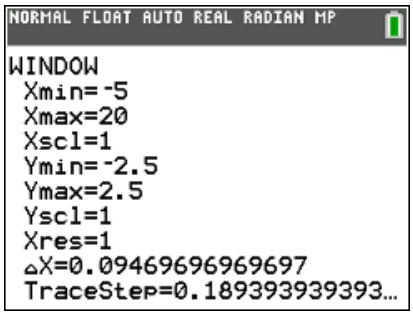
$$7\alpha^2 - 108\alpha - 244 = 0$$

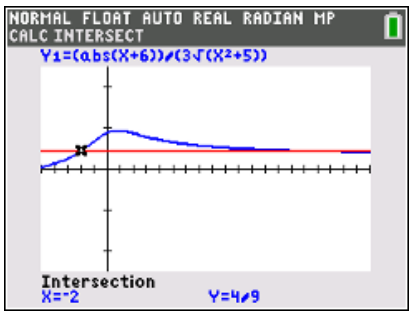
$$(\alpha+2)(7\alpha-122) = 0$$

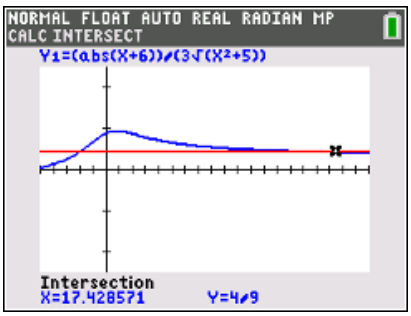
$$\alpha = -2 \text{ or } \alpha = \frac{122}{7}$$

Alternative method (using GC)









$\alpha = -2$ or $\alpha = 17.4$ (3 s.f)

(b)

$$p_1 : \mathbf{r} \bullet \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 5$$

$$l_1 : \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$$

Subst l_1 into p_1,

$$\begin{pmatrix} 1-2\lambda \\ 2-2\lambda \\ -3+\lambda \end{pmatrix} \bullet \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 5$$

$$-9+4\lambda = 5$$

$$\lambda = \frac{7}{2}$$

$$\overrightarrow{OP} = \begin{pmatrix} 1-2\left(\frac{7}{2}\right) \\ 2-2\left(\frac{7}{2}\right) \\ -3+\left(\frac{7}{2}\right) \end{pmatrix} = \begin{pmatrix} -6 \\ -5 \\ \frac{1}{2} \end{pmatrix}$$

(c)

Since B is on l_1, $\overrightarrow{OB} = \begin{pmatrix} 1-2\lambda \\ 2-2\lambda \\ -3+\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$.

$$\overrightarrow{AB} = \begin{pmatrix} 1-2\lambda \\ 2-2\lambda \\ -3+\lambda \end{pmatrix} - \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2-2\lambda \\ 1-2\lambda \\ -7+\lambda \end{pmatrix}$$

Since l_2 is parallel to p_1, $\overrightarrow{AB} \perp \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$

$$\begin{pmatrix} -2-2\lambda \\ 1-2\lambda \\ -7+\lambda \end{pmatrix} \bullet \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} = 0$$

$$\lambda = \frac{9}{2}$$

$$\overrightarrow{OB} = \begin{pmatrix} 1-2\left(\frac{9}{2}\right) \\ 2-2\left(\frac{9}{2}\right) \\ -7+\left(\frac{9}{2}\right) \end{pmatrix} = \begin{pmatrix} -8 \\ -7 \\ \frac{3}{2} \end{pmatrix}$$

The coordinates of B are $\left(-8, -7, \frac{3}{2}\right)$.

(d)	The direction vector of l_2 is parallel to $\overrightarrow{AB} = \begin{pmatrix} -2-2\left(\frac{9}{2}\right) \\ 1-2\left(\frac{9}{2}\right) \\ -7+\left(\frac{9}{2}\right) \end{pmatrix} = \begin{pmatrix} -11 \\ -8 \\ -\frac{5}{2} \end{pmatrix}$ $\cos\theta = \dfrac{\left\lvert \begin{pmatrix} -2 \\ -2 \\ 1 \end{pmatrix} \bullet \begin{pmatrix} -11 \\ -8 \\ -\frac{5}{2} \end{pmatrix} \right\rvert}{\sqrt{9}\sqrt{\frac{765}{4}}}$ $\theta = 31.166 \simeq 31.2^{\circ}$ (to 1 d.p.)
(e)	Point S must be on l_1 since every possible orientation of plane p_2 contains the point S. $\overrightarrow{OS} = \begin{pmatrix} 1-2\lambda \\ 2-2\lambda \\ -3+\lambda \end{pmatrix}$ for some $\lambda \in \mathbb{R}$. $\begin{pmatrix} 1-2\lambda \\ 2-2\lambda \\ -3+\lambda \end{pmatrix} = \begin{pmatrix} -3 \\ y \\ -1 \end{pmatrix}$ $1-2\lambda = -3 \Rightarrow \lambda = 2$ (**Alternatively** : using $z-$component to solve for $\lambda, -3+\lambda = -1 \Rightarrow \lambda = 2$) $2-2(2) = y \Rightarrow y = -2$